Vaishnavi Arruru

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.55
Sri Chaitanya junior kalashala	Class XII (Telangana state board of Intermediate education)	2023	98.20%
Sri Chaitanya High School	Class X (Board of Secondary Education)	2021	10.00%

WORK EXPERIENCE

Jio Platforms Limited AI Intern - Speech-to-Text for Indic Languages

-Contributed to the development and testing of multilingual **Speech-to-Text** (STT) systems supporting speech automation across **Jio platform** applications for **3** Indic languages.

-Built and evaluated end-to-end **STT** training pipelines by preprocessing and validating **3,000+** speech samples with aligned audio-text pairs for **model training** and **testing**.

-Developed a **code-switch-**aware **Speech-to-Text (STT) evaluation** and **normalization** pipeline for English, Hindi, and Telugu, achieving a **19.1% reduction in overall WER** and **22.9% reduction in overall CER** for Jio-domain and script-level errors.

SKILLS

-Languages: C++, C, Python, HTML, SQL, LaTeX

-Tech & Tools: FastAPI, React.js, Node.js, TypeScript, REST APIs, WebSockets, OOP, LangChain, LangGraph, Postman

-AI & ML: Deep Learning (PyTorch, CNN), Machine Learning, NLP (spaCy, Transformers), Speech-to-Text (STT), ASR, OpenCV, Random Forest, Geospatial ML

-DBMS: PostgreSQL, MySQL

-DevOps & OS: Docker, Git, Hugging Face, Linux, Windows

-Soft Skills: Communication, Problem-Solving, Adaptability, Agile

PROJECTS

-Crypto Live Dashboard

1)Built a **full-stack web app** that collects real-time crypto token data from **multiple APIs** and displays it in an interactive dashboard.

2)Created a **Node.js/TypeScript backend** with **WebSocket** updates to show live changes in price, volume, and trends.

3)Developed a responsive **React frontend** with **filtering** ,**sorting**, and **pagination** to let users easily browse and track live token data

-Satellite Temperature Gap Filling System

1)Developed a scalable **ML pipeline** for reconstructing missing Land Surface Temperature (LST) values using **25 years** (2001-2025) of real-world **MODIS satellite data** comprising **~9,100 daily thermal grids** at **1 km spatial resolution**.

2)Built a **Random Forest residual-learning model** using **17 spatio-temporal** and **topographic features** optimized

for large-scale geospatial datasets.

3) Achieved **35-46% lower RMSE** than classical interpolation methods on a strict 5-year (2021-2025) held-out benchmark, maintaining **$R^2 > 0.84$** even under **30% artificial gap density**

-AI-Powered Document Processing System

1)Built a document classification module using **CNN/ResNet** to categorize invoices, resumes, and contracts with high accuracy.

2)Designed an **AI agent pipeline** with **NLP models** (**spaCy, Transformers**) to extract key fields such as names, dates, and amounts.

3)Deployed the system as a **RESTful API** (FastAPI) with **Docker** .

POSITIONS OF RESPONSIBILITY

-Manager and **Assistant Manager** in **Ecell IITH** Ideation domain

-Core Member of **Media Council**, **Art Club**, and **Filmmaking Club** of IITH

-Coordinator in **Web** Domain for **Diesta 2026**

-Coordinator in **INTER IIT Sports Meet** (Events)

-Buddie in **Sunshine**-The Counselling Cell IITH

SCHOLASTIC ACHIEVEMENTS

-Selected in **National Talent Search Examination** Level-1 conducted by NCERT

-Ranked **16th** in 50th National Mathematics Talent Contest(NMTC) among the top scorers in **Sub-Junior**

Level final test held by Associates of Mathematics Teachers of India (**AMTI**)

RELEVANT COURSES

-Introduction to Programming by Professor Vineeth N Balasubramanian and Mourendra

-Introduction to SQL with MySQL- Udemy

-Data Structures and Algorithms

EXTRACURRICULAR

-Participated in **Carrom** competitions and **Cricket** championship(runnerups) during Diesta fest

-Participated in **Solo dance** competition during **Diesta** fest